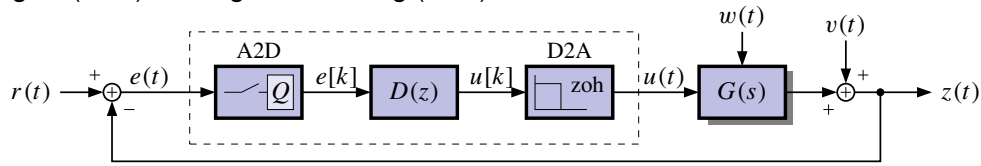




## Discrete-time state-space models

- Computer monitoring of real-time systems requires analog-to-digital (A2D) and digital-to-analog (D2A) conversion.



- Linear discrete-time systems can also be represented in state-space form:

$$\begin{aligned} x_{k+1} &= A_d x_k + B_d u_k + w_k \\ z_k &= C_d x_k + D_d u_k + v_k. \end{aligned}$$

- The subscript “d” emphasizes that, in general, the “A”, “B”, “C” and “D” matrices are different for the same discrete-time and continuous-time system.
- I will usually drop the “d” and expect you to interpret the system from its context.



## Time (dynamic) response of discrete-time models

- The full solution, via induction from  $x_{k+1} = A_d x_k + B_d u_k$ , is:

$$x_k = A_d^k x_0 + \underbrace{\sum_{j=0}^{k-1} A_d^{k-1-j} B_d u_j}_{\text{convolution}}.$$

- Since  $z_k = C_d x_k + D_d u_k$ , we also have:

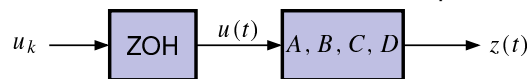
$$z_k = \underbrace{C_d A_d^k x_0}_{\text{initial resp.}} + \underbrace{\sum_{j=0}^{k-1} C_d A_d^{k-1-j} B_d u_j}_{\text{convolution}} + \underbrace{D_d u_k}_{\text{feedthrough}}.$$

- Comparing with the continuous-time solution,  $e^{At}$  has been replaced by  $A_d^k$  and integrals have been replaced by summations.



## Converting plant dynamics to discrete time (1)

- Combine the dynamics of the zero-order hold and the plant.



- Recall that the continuous-time state dynamics of the plant are:

$$\dot{x}(t) = Ax(t) + Bu(t).$$

- Evaluate  $x(t)$  at discrete times. Recall also the solution for  $x(t)$ :<sup>1</sup>

$$\begin{aligned} x(t) &= \int_0^t e^{A(t-\tau)} Bu(\tau) d\tau \\ x_{k+1} &= x((k+1)\Delta t) = \int_0^{(k+1)\Delta t} e^{A((k+1)\Delta t-\tau)} Bu(\tau) d\tau. \end{aligned}$$

<sup>1</sup>For simplicity, we assume that  $x(0) = 0$ . The final solution turns out to be true even if  $x(0) \neq 0$ .



## Converting plant dynamics to discrete time (2)

- We break up the integral into two pieces:

$$\begin{aligned}
 x_{k+1} &= \int_0^{(k+1)\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau \\
 &= \int_0^{k\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau + \int_{k\Delta t}^{(k+1)\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau \\
 &= \int_0^{k\Delta t} e^{A\Delta t} e^{A(k\Delta t - \tau)} B u(\tau) d\tau + \int_{k\Delta t}^{(k+1)\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau \\
 &= e^{A\Delta t} x(k\Delta t) + \int_{k\Delta t}^{(k+1)\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau.
 \end{aligned}$$

- The first integral has become  $A_d$  times  $x(k\Delta t)$ .
- The second will become  $B_d$  times  $u(k\Delta t)$  after a little more work.



## Converting plant dynamics to discrete time (3)

- In the remaining integral, note that  $u(\tau)$  is assumed to be constant from  $k\Delta t$  to  $(k+1)\Delta t$ , and equal to  $u(k\Delta t)$ .
- We evaluate the integral via change of variables. Let  $\sigma = (k+1)\Delta t - \tau$ ;  $\tau = (k+1)\Delta t - \sigma$ ;  $d\tau = -d\sigma$ .

$$\begin{aligned}
 x_{k+1} &= e^{A\Delta t} x(k\Delta t) + \int_{k\Delta t}^{(k+1)\Delta t} e^{A((k+1)\Delta t - \tau)} B u(\tau) d\tau \\
 &= e^{A\Delta t} x(k\Delta t) + \left[ \int_0^{\Delta t} e^{A\sigma} B d\sigma \right] u(k\Delta t) \\
 &= \underbrace{e^{A\Delta t}}_{A_d} x_k + \underbrace{\left[ \int_0^{\Delta t} e^{A\sigma} B d\sigma \right]}_{B_d} u_k.
 \end{aligned}$$



## Computing $A_d$ , $C_d$ , and $D_d$

- Summarizing to this point, we have a discrete-time state-space representation from the continuous-time representation:

$$x_{k+1} = A_d x_k + B_d u_k,$$

where  $A_d = e^{A\Delta t}$  and  $B_d = \int_0^{\Delta t} e^{A\sigma} B d\sigma$ .

- Similarly, we have:

$$z_k = C x_k + D u_k.$$

where  $C_d = C$  and  $D_d = D$ .

- So, there is no conversion for the  $C$  and  $D$  matrices, and the conversion for  $A$  is straightforward via the matrix exponential  $A_d = e^{A\Delta t}$ . This is different from taking the exponential of each element in  $A\Delta t$ .
- If Octave is handy, you can type in:  $A_d = \text{expm}(A * \Delta t)$ . Otherwise, you will need to compute  $e^{A\Delta t} = \mathcal{L}^{-1}[(sI - A)^{-1}]|_{t=\Delta t}$ .



## Computing $B_d$

- Now we focus on computing  $B_d$ . Recall that

$$\begin{aligned}
 B_d &= \int_0^{\Delta t} e^{A\sigma} B \, d\sigma \\
 &= \int_0^{\Delta t} \left( I + A\sigma + A^2 \frac{\sigma^2}{2} + \dots \right) B \, d\sigma \\
 &= \left( I\Delta t + A \frac{\Delta t^2}{2!} + A^2 \frac{\Delta t^3}{3!} + \dots \right) B \\
 &= A^{-1}(e^{A\Delta t} - I)B = A^{-1}(A_d - I)B.
 \end{aligned}$$

- If  $A$  is invertible, this method works nicely; otherwise, we will need to perform the integral in the first line manually.
- Also, in Octave, `[Ad,Bd]=c2d(A,B,dT)`.



## Summary

- Computer monitoring of physical systems requires A2D and D2A conversion.
- The system that is “seen” by the computer is a discrete-time system, and must be represented by a discrete-time state-space model:

$$\begin{aligned}
 x_{k+1} &= A_d x_k + B_d u_k + w_k \\
 z_k &= C_d x_k + D_d u_k + v_k.
 \end{aligned}$$

- In general, the “ $A$ ”, “ $B$ ”, “ $C$ ” and “ $D$ ” matrices are different for the same discrete-time and continuous-time system.
- You have learned how to convert from the original “ $A$ ”, “ $B$ ”, “ $C$ ” and “ $D$ ” matrices to the discrete-time “ $A_d$ ”, “ $B_d$ ”, “ $C_d$ ” and “ $D_d$ ” matrices.
- In the next lesson, you will see some examples of converting continuous-time to discrete-time and of simulating discrete-time state-space models.